

DAVID EZEKIEL LUTTA

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PROFESSIONAL SUMMARY

Software Engineer II at Microsoft with 4+ years of experience building AI-powered and distributed systems at scale. Currently improving Microsoft Copilot's AI model quality and reliability across Microsoft 365; previously built low-latency search services used across the M365 product suite. Focused on LLM orchestration, prompt tuning, and practical AI systems from cloud-scale services to self-hosted edge inference.

SKILLS

- **AI & LLM Systems:** LLM Orchestration, Prompt Tuning & Evaluation, Multi-Agent Systems, OpenAI API, Computer Vision (YOLOv8)
- **Languages:** C#, Java, JavaScript, TypeScript, Python
- **Backend & Distributed Systems:** .NET, Node.js, API Design, Database Development, Design Patterns, Algorithm Analysis & Design
- **Databases:** SQL Server, MongoDB, Firebase
- **Cloud & DevOps:** Microsoft Azure, Azure DevOps (CI/CD), Docker

EXPERIENCE

Software Engineer II — Microsoft

Sep 2025 – Present

- Improve the quality, reliability, and performance of Microsoft Copilot by rolling out AI model upgrades, delivering faster and more accurate responses across Microsoft 365

Software Engineer — Microsoft

Aug 2022 – Sep 2025

- Contributed to low-latency search services for Microsoft 365 as part of the Microsoft AI (MSAI) team, optimizing query performance and response reliability across Microsoft products

Software Engineering Intern — Microsoft

Jun 2021 – Sep 2021

- Enhanced the relevance of people-search suggestions within Microsoft 365, helping users find the right people and content faster

HIGHLIGHTED PROJECTS

Utafiti Bot

Open-source multi-agent research orchestrator built in TypeScript, using the OpenAI API without third-party agent frameworks. Designed a typed message bus and orchestrator agent with fan-out/fan-in coordination across specialist agents (web search, data analysis, synthesis), plus a two-tier memory system for session and cross-session context.

Tech: TypeScript, OpenAI API, Brave Search API, lowdb, Chroma

Forest Guard

Real-time IoT and machine learning system for early detection of illegal logging and forest fires. Unlike satellite-based monitoring, Forest Guard detects activity on the ground as it happens.

Tech: Azure, C#, Python, C++

Smart Toll System

IoT-based project enabling drivers to pay toll fees via RFID tags and a smartphone, integrated with M-Pesa for digital payment. Built the RFID hardware layer, admin web dashboard, and driver mobile app end-to-end.

Tech: Raspberry Pi, RFID, Firebase, M-Pesa API, JavaScript

KodiPay

Full-stack application that automatically processes SMS rent payment notifications, extracts payment details via regex parsing, and syncs data to Google Sheets via API. Used MongoDB for persistent storage and JWT authentication for secure access to payment records and the tenant management dashboard.

Tech: Node.js, MongoDB, Google Sheets API, JWT, Azure

YTS Torrent Box

Application that displays movies from the YTS database; users can initiate downloads that run on a home Raspberry Pi server, with real-time progress synced back to mobile and web clients.

Tech: Android, Raspberry Pi, Firebase, YTS API

EDUCATION

Jomo Kenyatta University of Agriculture and Technology (JKUAT)

Bachelor of Science in Information Technology — Second Class Honors, Upper Division

AWARDS

- Microsoft Global Hackathon 2023 — Award Winner [confirm project name/month to strengthen this line]
- Microsoft Student Hackathon 2021 — Grand Prize Winner (Global): built "Forest Guard," an IoT and Azure-based system to reduce deforestation through real-time detection of illegal logging and forest fires

VOLUNTEER EXPERIENCE

Technical Mentor — Young Scientists Kenya

Mar 2018 – Aug 2018

- Mentored high school students on programming projects; one mentee group qualified for the final round

REFERENCES

Available upon request.